

BERTlab - German grammar

Thanks to the pretraining process, multilingual BERT has a good command of German grammar out of the box

- ▶ BERT has cases and genders figured out

```
Wir gehen durch _ Wald.  
[('den', 0.8579956), ('einen', 0.054591507), ('ein', 0.016730022), ('diesen', 0.015647054)]
```

- ▶ Excellent job with articles, case, and gender here

```
Ich träume von _ sprechenden Delphin.  
[('einem', 0.6294187), ('dem', 0.20742233), ('einer', 0.049603593), ('der', 0.036961004)]
```

- ▶ BERT remains consistent if we change the sentence

```
Ich träume davon, mit _ Delfin zu sprechen.  
[('dem', 0.39727402), ('einem', 0.33334878), ('der', 0.13823836), ('einer', 0.058449622), ('den', 0.012854406),
```

Give BERT some harder examples and see what happens!

BERTlab - Coreference Resolution

Recall Winograd schemas from the Information Extraction lecture.
Here's a well-known example:

The trophy didn't fit in the suitcase because it was too big/small.

- ▶ As a human, you know that *it* refers to the trophy if it is too big, and the suitcase if it is too small.
- ▶ How do we find out whether Bert knows that?
- ▶ Get BERT to pick the correct pronouns in German!
- ▶ If we make the suitcase too small, BERT picks the correct pronoun. How confident would you say BERT is?

```
Die Trophäe würde nicht in den braunen Koffer passen, weil _ zu klein ist.  
[('er', 0.3036696), ('sie', 0.29711017), ('es', 0.19701958), ('diese', 0.037451033),
```

- ▶ If we make the trophy too big, BERT chooses incorrectly.

```
Die Trophäe würde nicht in den braunen Koffer passen, weil _ zu gross ist.  
[('er', 0.3046717), ('es', 0.2720747), ('sie', 0.21926354), ('diese', 0.023713931),
```

BERTlab - Coreference Resolution

Let's try with English: *Mark visited Janet's grave in 1765. At that time, he/she had been traveling/dead for five years.*

- ▶ As a human, you know that *he* goes with *traveling* and *she* goes with *dead*
- ▶ BERT seems rather confident that Janet had been dead

```
Mark visited Janet's grave in 1765. At that time, _ had been dead for five years.
```

```
[('Janet', 0.22340725), ('she', 0.19979894), ('Margaret', 0.028870035), ('he', 0.028126137), ('John', 0.014088499),
```

- ▶ BERT seems confused about whether she was also traveling (look at how close the probabilities are!).

```
Mark visited Janet's grave in 1765. At that time, _ had been traveling for five years.
```

```
[('she', 0.12657051), ('he', 0.11243445), ('they', 0.05081436), ('Janet', 0.023389611), ('John', 0.018392442),
```

Let's be fair to BERT: we're using it out of the box to solve a task it should be fine-tuned for!

BERTlab - Cloze Test

A cloze test requires examinees to fill in missing words (or sentences) to best fit the surrounding context

Passage: Nancy had just got a job as a secretary in a company. Monday was the first day she went to work, so she was very .1. and arrived early. She .2. the door open and found nobody there. "I am the .3. to arrive." She thought and came to her desk. She was surprised to find a bunch of .4. on it. They were fresh. She .5. them and they were sweet. She looked around for a .6. to put them in. "Somebody has sent me flowers the very first day!" she thought .7. . " But who could it be?" she began to .8. . The day passed quickly and Nancy did everything with .9. interest. For the following days of the .10. , the first thing Nancy did was to change water for the followers and then set about her work.

Then came another Monday. .11. she came near her desk she was overjoyed to see a(n) .12. bunch of flowers there. She quickly put them in the vase, .13. the old ones. The same thing happened again the next Monday. Nancy began to think of ways to find out the .14. . On Tuesday afternoon, she was sent to hand in a plan to the .15. . She waited for his directives at his secretary's .16. . She happened to see on the desk a half-opened notebook, which .17. : "In order to keep the secretaries in high spirits, the company has decided that every Monday morning a bunch of fresh flowers should be put on each secretaries desk." Later, she was told that their general manager was a business management psychologist.

Questions:

- | | | | |
|----------------------|------------------|---------------------|-------------------|
| 1. A. depressed | B. encouraged | C. excited | D. surprised |
| 2. A. turned | B. pushed | C. knocked | D. forced |
| 3. A. last | B. second | C. third | D. first |
| 4. A. keys | B. grapes | C. flowers | D. bananas |
| 5. A. smelled | B. ate | C. took | D. held |
| 6. A. vase | B. room | C. glass | D. bottle |
| 7. A. angrily | B. quietly | C. strangely | D. happily |
| 8. A. seek | B. wonder | C. work | D. ask |
| 9. A. low | B. little | C. great | D. general |
| 10. A. month | B. period | C. year | D. week |
| 11. A. Unless | B. When | C. Since | D. Before |
| 12. A. old | B. red | C. blue | D. new |
| 13. A. covering | B. demanding | C. replacing | D. forbidding |
| 14. A. sender | B. receiver | C. secretary | D. waiter |
| 15. A. assistant | B. colleague | C. employee | D. manager |
| 16. A. notebook | B. desk | C. office | D. house |
| 17. A. said | B. written | C. printed | D. signed |

Figure: *

Source: Q. Xie et al.,
Large-scale Cloze Test Dataset
Created by Teachers, EMNLP
2018

BERTlab - Cloze Test

BERT does very well with the first blank: the correct answer, *excited* is chosen with probability 0.85

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Switching to bert-large-uncased
Nancy had just got a new job in a company. Monday was the first day she went to work, so she was very _ and
[('excited', 0.85088223), ('surprised', 0.13961668), ('encouraged', 0.0048667174), ('depressed', 0.004634448)]
The most likely word is "excited".
```

BERT doesn't do as well with the second blank: nobody would *turn* a door open in English!

```
She _ the door open and found nobody there.
[('turned', 0.5521097), ('pushed', 0.22897945), ('knocked', 0.19420265), ('forced', 0.024708237)]
The most likely word is "turned".
```

But wait, we're not being fair. If we give BERT the whole sentence up to that point, BERT gives us the correct answer.

```
Nancy had just got a new job in a company. Monday was the first day she went to work, so she was very excited and arrived early
She _ the door open and found nobody there.
[('pushed', 0.8576227), ('forced', 0.079986386), ('turned', 0.046390593), ('knocked', 0.016000306)]
The most likely word is "pushed".
```

BERTlab - Cloze Test

BERT also gives the correct answer for the blank in the sentence *I am the _ to arrive.*

Nancy had just got a new job in a company. Monday was the first day she went to work, so she was very excited and arrived early. She pushed the door open and found nobody there.

"I am the _ to arrive." She thought and came to her desk.

```
[('first', 0.6768758), ('last', 0.23204431), ('second', 0.07630599), ('third', 0.014773911)]
```

The most likely word is "first".

We don't even need to give BERT the whole text from the very beginning:

She pushed the door open and found nobody there. "I am the _ to arrive." She thought and came to her desk.

```
[('first', 0.86124), ('second', 0.073761046), ('last', 0.040146008), ('third', 0.02485288)]
```

The most likely word is "first".

BERT only needs one context sentence to get *flowers* right in the next blank.

"I am the first to arrive." She thought and came to her desk. She was surprised to find a bunch of _ on it.

```
[('flowers', 0.9084021), ('keys', 0.06575211), ('grapes', 0.015442871), ('bananas', 0.010403007)]
```

The most likely word is "flowers".

BERTlab - Cloze Test

The next blank is very hard to guess and BERT barely gets it right

```
She was surprised to find a bunch of flowers on it. They were fresh. She _ them and they were sweet. She looked around for  
[('smelled', 0.32999295), ('took', 0.31739318), ('held', 0.19081923), ('ate', 0.16179474)]  
The most likely word is "smelled".
```

But BERT can easily look farther back in the text and increase the margin of confidence of the prediction.

```
"I am the first to arrive." She thought and came to her desk. She was surprised to find a bunch of flowers on it. They were  
[('smelled', 0.6510981), ('ate', 0.15703304), ('took', 0.12225221), ('held', 0.069616765)]  
The most likely word is "smelled".
```

This is one of the examples reported as problematic in (Xie et al., 2018)¹

BERT can do all that with no task-specific fine-tuning!

¹Q. Xie et al., Large-scale Cloze Test Dataset Created by Teachers, EMNLP 2018